TM Forum Specification

Document Management User Guide

TMF667

Maturity Level: General Availability (GA)	Team Approved Date: 04-Aug-2026
Release Status: Production	Approval status: Team Approved
Version: 5.0.0	IPR Mode: RAND

Notice

Copyright © TM Forum 2026. All Rights Reserved.

This document and translations of it may be copied and furnished to others, and derivative works that comment on or otherwise explain it or assist in its implementation may be prepared, copied, published, and distributed, in whole or in part, without restriction of any kind, provided that the above copyright notice and this section are included on all such copies and derivative works. However, this document itself may not be modified in any way, including by removing the copyright notice or references to TM FORUM, except as needed for the purpose of developing any document or deliverable produced by a TM FORUM Collaboration Project Team (in which case the rules applicable to copyrights, as set forth in the TM FORUM IPR Policy, must be followed) or as required to translate it into languages other than English.

The limited permissions granted above are perpetual and will not be revoked by TM FORUM or its successors or assigns.

This document and the information contained herein is provided on an "AS IS" basis and TM FORUM DISCLAIMS ALL WARRANTIES, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTY THAT THE USE OF THE INFORMATION HEREIN WILL NOT INFRINGE ANY OWNERSHIP RIGHTS OR ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

TM FORUM invites any TM FORUM Member or any other party that believes it has patent claims that would necessarily be infringed by implementations of this TM Forum Standards Final Deliverable, to notify the TM FORUM Team Administrator and provide an indication of its willingness to grant patent licenses to such patent claims in a manner consistent with the IPR Mode of the TM FORUM Collaboration Project Team that produced this deliverable.

TM FORUM invites any party to contact the TM FORUM Team Administrator if it is aware of a claim of ownership of any patent claims that would necessarily be infringed by implementations of this TM FORUM Standards Final Deliverable by a patent holder that is not willing to provide a license to such patent claims in a manner consistent with the IPR Mode of the TM FORUM Collaboration Project Team that produced this TM FORUM Standards Final Deliverable. TM FORUM may include such claims on its website but disclaims any obligation to do so.

TM FORUM takes no position regarding the validity or scope of any intellectual property or other rights that might be claimed to pertain to the implementation or use of the technology described in this TM FORUM Standards Final Deliverable or the extent to which any license under such rights might or might not be available; neither does it represent that it has made any effort to identify any such rights. Information on TM FORUM's procedures

with respect to rights in any document or deliverable produced by a TM FORUM Collaboration Project Team can be found on the TM FORUM website. Copies of claims of rights made available for publication and any assurances of licenses to be made available, or the result of an attempt made to obtain a general license or permission for the use of such proprietary rights by implementers or users of this TM FORUM Standards Final Deliverable, can be obtained from the TM FORUM Team Administrator. TM FORUM makes no representation that any information or list of intellectual property rights will at any time be complete, or that any claims in such list are, in fact, Essential Claims.

Direct inquiries to the TM Forum office:

100 Enterprise Drive
Suite 301 #1649
Rockaway, NJ 070866, USA
Tel No. +1 862 227 1648
TM Forum Web Page: www.tmforum.org

© TM Forum 2026. All Rights Reserved.

Table of Contents

Notice . . . i
Introduction . . . 1
Document . . . 1
Document Specification . . . 1
Attachment . . . 1
Sample Use Cases . . . 3
Retrieve all the documents stored for a Customer . . . 3
Register Customer Identification Document during order capture . . . 3
Support of polymorphism and extension patterns . . . 5
RESOURCE MODEL . . . 6
Managed Entity and Task Resource Models . . . 6
Document resource . . . 6
DocumentSpecification resource . . . 24
Attachment resource . . . 48
Notification Resource Models . . . 50
Document . . . 52
DocumentSpecification . . . 55
Attachment . . . 58
API OPERATIONS . . . 60
Operations on Document . . . 60
Retrieves a document by ID . . . 60
List or find document objects . . . 62
Creates a document . . . 64
Updates partially a document . . . 67
Deletes a document . . . 76
Operations on DocumentSpecification . . . 76
Retrieves a documentSpecification by ID . . . 76
List or find documentSpecification objects . . . 78
Creates a documentSpecification . . . 80
Updates partially a documentSpecification . . . 82
Deletes a documentSpecification . . . 90
Operations on Attachment . . . 90
Retrieves a attachment by ID . . . 90

© TM Forum 2026. All Rights Reserved.

List or find attachment objects . . . 91
Creates a attachment . . . 92
Updates partially a attachment . . . 93
Deletes a attachment . . . 97
API NOTIFICATIONS . . . 98
Register listener . . . 98
Unregister listener . . . 99
Publish Event to listener . . . 99
Acknowledgements . . . 101
Release History . . . 101
Contributors to Document . . . 101

© TM Forum 2026. All Rights Reserved.

Introduction

The following document is the user guide of the REST API for Document, Document Specification, and Attachment management. It provides a comprehensive definition of the resource models and details all available operations

Document

The Document resource manages the metadata associated with a record. This includes administrative data such as related parties (e.g., author, reviewer, publisher), creation dates, versioning, and status.

The Document does not contain the binary content itself; rather, the actual "body" of the document is managed or referenced by an associated Attachment. This allows the metadata to remain lightweight while the content, whether encoded directly as binary data or provided via a reference (e.g., https: or file:), is handled separately.

Document Specification

The DocumentSpecification resource follows the EntitySpecification pattern, serving as a standardized blueprint for creating specific document categories. It includes attributes for lifecycle management, versioning, and relationships, alongside the document type and its characteristics specifications. Each Document type (such as book, commercial agreement, passport, articles of incorporation, training video) might be described by its own Document Specification. This resource manages the collection of characteristics that detail the underlying media type and its metadata. For example:

- A Document Specification for “book” might have characteristics such as `pages` (page count), `language`, `isbn-10`, `publisher`, and physical dimensions.
- A Document Specification for “training video” might have characteristics such as `objective` (learning objective), `duration`, `difficulty` (difficulty level), `transcript` (transcript available), and `resolution`.

The specification allows you to define both the structure of the characteristics and the constraints of their values.

Attachment

The Attachment represents the digital or physical realization of the document. While the Document instance manages the "data about the data," the Attachment provides the link to the content itself via the `url` attribute or contains the raw data in the `content` attribute. Additionally in Version 5.0.0 the Attachment was added with an Endpoint to the Document

© TM Forum 2026. All Rights Reserved.

Management API. Many API models use Attachment as a resource directly, without needing the full complexity and business meaning of Document.

© TM Forum 2026. All Rights Reserved.

Sample Use Cases

Retrieve all the documents stored for a Customer

The Document Management API can be used to retrieve all the identification documents stored for a Customer in a Document Management system.

The following request retrieves all identification documents linked to the customer whose party role has id `b7f2-c3d4e5f60789`:

```
GET /tmf-api/document/v5/document?relatedParty.partyOrPartyRole.id=b7f2-
c3d4e5f60789&relatedParty.partyOrPartyRole.@type=PartyRoleRef&documentType=identificati
onDocument
```

Register Customer Identification Document during order capture

The Document Management API can be used to register receipt of a customer identification document during a Product Order capture flow.

The following request registers a passport scan for customer `b7f2-c3d4e5f60789`, associating it with product order `f1e2-d3c4b5a60987`:

```
POST /tmf-api/document/v5/document
```

```
{
  "@type": "Document",
  "name": "Jack Smith - Passport",
  "documentType": "identificationDocument",
  "status": "received",
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Customer",
      "partyOrPartyRole": {
        "@type": "PartyRoleRef",
        "href": "https://serverRoot/tmf-api/partyRoleManagement/v5/partyRole/b7f2-
c3d4e5f60789",
        "id": "b7f2-c3d4e5f60789",
        "name": "Jack Smith",
        "@referredType": "PartyRole"
      }
    }
  ],
  "relatedEntity": [
    {
      "@type": "RelatedEntity",
      "role": "relatedOrder",
```

© TM Forum 2026. All Rights Reserved.

```
      "entity": {
        "@type": "EntityRef",
        "@referredType": "ProductOrder",
        "href": "https://serverRoot/tmf-
api/productOrderingManagement/v5/productOrder/f1e2-d3c4b5a60987",
        "id": "f1e2-d3c4b5a60987"
      }
    }
  ],
  "attachment": [
    {
      "@type": "Attachment",
      "name": "Passport scan",
      "url": "https://dms.example.com/docs/passport-john-doe.pdf",
      "attachmentType": "Passport",
      "mimeType": "application/pdf"
    }
  ]
}
```


© TM Forum 2026. All Rights Reserved.

Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a BillingAccount and SettlementAccount inheriting properties from the abstract Account entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Account instances some may be instances of BillingAccount where other could be instances of SettlementAccount. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an AccountRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as BillingAccountRef or SettlementAccountRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

© TM Forum 2026. All Rights Reserved.

RESOURCE MODEL

Managed Entity and Task Resource Models

Document resource

Manages document metadata, status, and versioning, including related parties like authors or publishers. Content is handled via an Attachment as either inline binary (content) or an external reference (url). Detailed attributes for specific document types are governed by a linked DocumentSpecification.

Resource model

© TM Forum 2026. All Rights Reserved.

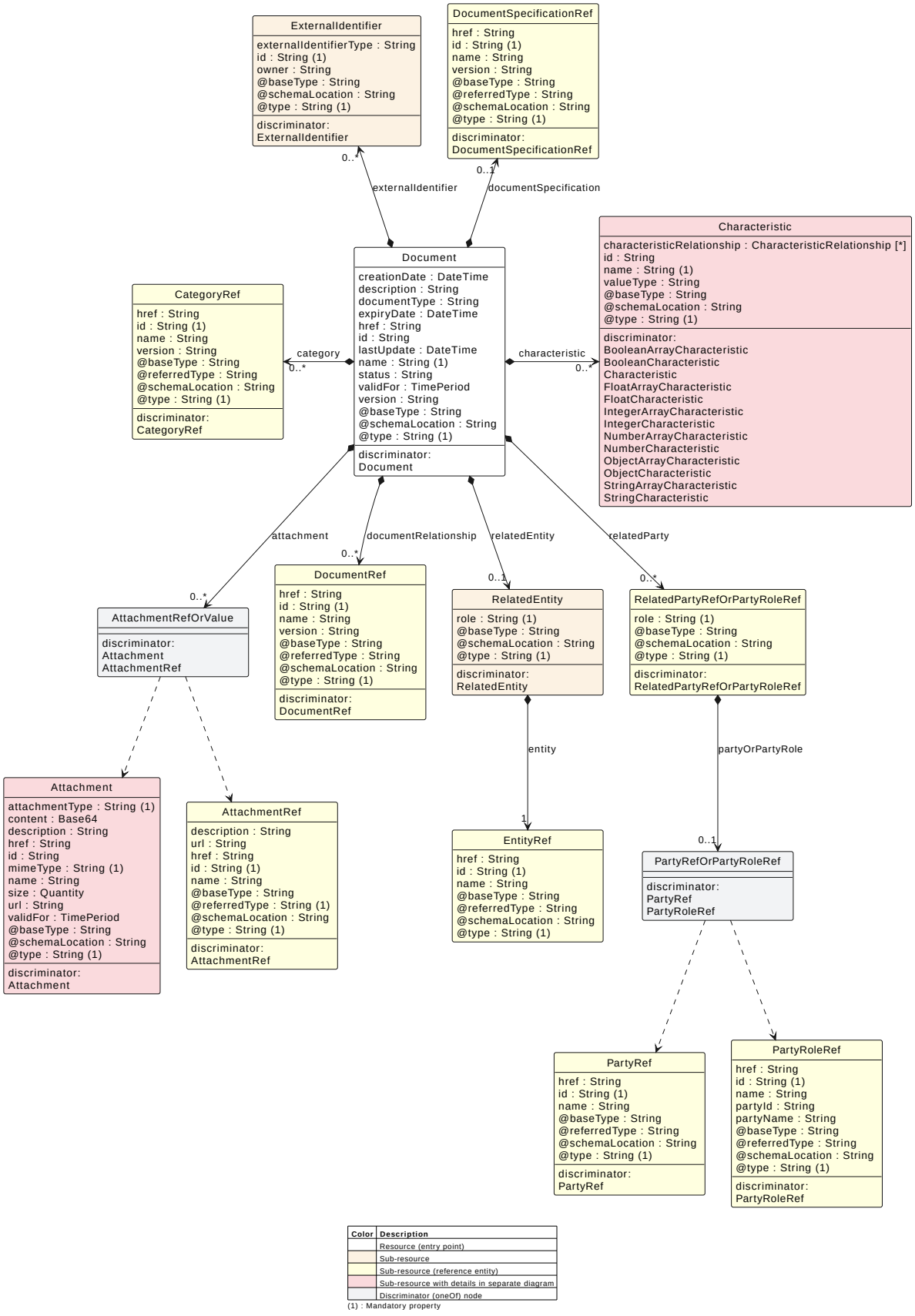


Figure 1 - Document

© TM Forum 2026. All Rights Reserved.

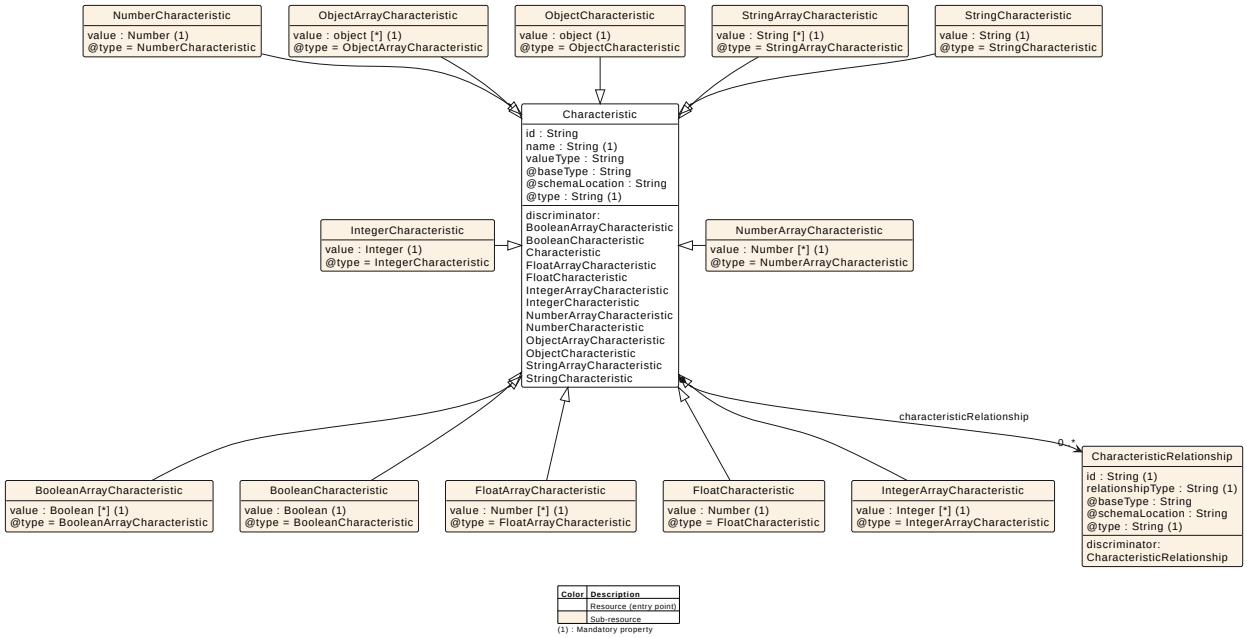


Figure 2 - Characteristic

Field descriptions

Document fields

attachment	An AttachmentRefOrValue [0..*]. The actual content of the document, provided either by value (inline data or a URL to the content) or by reference to an existing Attachment resource. A document may have multiple attachments representing different components of the same content.
~~category~~	DEPRECATED A CategoryRef [0..*]. Collection of categories applicable.

© TM Forum 2026. All Rights Reserved.

characteristic	A Characteristic [0..*]. A collection of name-value pairs that describe additional properties of the document. The set of permitted characteristics and their value types are defined by the associated DocumentSpecification (via the specCharacteristic list). For example, a document based on a 'paperback' specification may carry characteristics such as pages (integer, e.g. 312), language (string, e.g. 'en'), or isbn-10 (string, e.g. '9781492026921'). Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
creationDate	A DateTime. The date and time the document was created. A date and time formatted in compliance with the ISO8601 standard must be used. To be set by the API server.
description	A String. Free-text description of the document.
documentRelationship	A DocumentRef [0..*]. A collection of references to documents that are related to this document. Documents may be related for various business reasons, such as a contract amendment superseding an original contract, an invoice supported by an itemized bill, or a legal attachment forming part of an agreement.
documentSpecification	A DocumentSpecificationRef. Reference to a DocumentSpecification entity.
documentType	A String. Name of the document type, for example: paperback, passport.

© TM Forum 2026. All Rights Reserved.

expiryDate	A DateTime. The date and time the document is to expire. Unlike validFor.endDateTime, which marks the end of the document's legal or commercial effectiveness, this field refers to the administrative expiry of the document. A date and time formatted in compliance with the ISO8601 standard must be used.
externalIdentifier	An ExternalIdentifier [0..*]. Identifiers assigned to this document by external systems (e.g. a KYC platform or document archive), allowing the document to be correlated across systems.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
lastUpdate	A DateTime. The date and time the document was last modified. A date and time formatted in compliance with the ISO8601 standard must be used. To be set by the API server.
name	A String [1]. A string used to give a name to the document.
relatedEntity	A RelatedEntity. A reference to an entity, where the type of the entity is not known in advance.
relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. A collection of related parties.
status	A String. Represents the status of the document. Example: draft, proposed, approved, published, cancelled, superseded.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. The version of the document.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Attachment sub-resource fields

Complements the description of an element (for instance a product) through video, pictures...

© TM Forum 2026. All Rights Reserved.

attachmentType	A String [1]. A business characterization of the purpose of the attachment, for example logo, instructionManual, contractCopy.
content	A Base64. The actual contents of the attachment object, embedded inline and encoded as base64. Mutually exclusive with url.
description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
mimeType	A String [1]. A technical characterization of the attachment content format using an IETF Media Type, for example application/pdf or image/png. Note: this property is named mimeType for historical reasons.
name	A String. The name of the attachment.
size	A Quantity. An amount in a given unit.
url	A String. The URL pointing to the attachment content held in an external system, such as a content management system or document repository. Mutually exclusive with content.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

AttachmentRef sub-resource fields

Attachment reference. An attachment complements the description of an element (for instance a product) through video, pictures.

~~description~~	DEPRECATED A String. A narrative text describing the content of the attachment. DEPRECATED: will be removed in v6.

© TM Forum 2026. All Rights Reserved.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
~~url~~	DEPRECATED A String. Link to the attachment media/content. DEPRECATED: will be removed in v6.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String [1]. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

CategoryRef sub-resource fields

Reference to a category in the catalog.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
version	A String. Version of the category.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : String Characteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, Number Characteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic...

© TM Forum 2026. All Rights Reserved.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
value	This property is present in subclasses

CharacteristicRelationship sub-resource fields

Another Characteristic that is related to the current Characteristic;.

id	A String [1]. Unique identifier of the characteristic.
relationshipType	A String [1]. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

© TM Forum 2026. All Rights Reserved.

BooleanArrayCharacteristic sub-resource fields

A characteristic which value is an array of Boolean(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Boolean [1..*]. A characteristic which value is an array of Boolean(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristic sub-resource fields

A characteristic which value is a Boolean.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Boolean [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FloatArrayCharacteristic sub-resource fields

A characteristic which value is an array of Float(s).

© TM Forum 2026. All Rights Reserved.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Float [1..*]. A characteristic which value is an array of Float(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FloatCharacteristic sub-resource fields

A characteristic which value is a Float.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Float [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristic sub-resource fields

A characteristic which value is an array of Integer(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.

© TM Forum 2026. All Rights Reserved.

name	A String [1]. Name of the characteristic.
value	An Integer [1..*]. A characteristic which value is an array of Integer(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristic sub-resource fields

A characteristic which value is a Integer.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	An Integer [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristic sub-resource fields

A characteristic which value is an array of Number(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Number [1..*]. A characteristic which value is an array of Number(s).

© TM Forum 2026. All Rights Reserved.

valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristic sub-resource fields

A characteristic which value is a Number.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Number [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristic sub-resource fields

A characteristic which value is an array of Object(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A valueType [1..*]. Collection of characteristic values.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2026. All Rights Reserved.

@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristic sub-resource fields

A characteristic which value is a Object.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A valueType_2 [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristic sub-resource fields

A characteristic which value is an array of String(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A String [1..*]. Collection of string characteristics.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristic sub-resource fields

© TM Forum 2026. All Rights Reserved.

A characteristic which value is a String.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A String [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

DocumentRef sub-resource fields

Reference to a Document entity.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
version	A String. Version of the document.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

DocumentSpecificationRef sub-resource fields

Reference to a DocumentSpecification entity.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.

© TM Forum 2026. All Rights Reserved.

name	A String. Name of the referred entity.
version	A String. Version of the document specification.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier sub-resource fields

An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.
id	A String [1]. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedEntity sub-resource fields

A reference to an entity, where the type of the entity is not known in advance.

© TM Forum 2026. All Rights Reserved.

entity	An EntityRef [1]. Entity reference schema to be use for all entityRef class.
role	A String [1].
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

EntityRef sub-resource fields

Entity reference schema to be use for all entityRef class.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.

partyOrPartyRole	A PartyRefOrPartyRoleRef.
role	A String [1]. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2026. All Rights Reserved.

@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

A Party reference.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. ENUMERATED with values: * PartyRef

PartyRoleRef sub-resource fields

Party role reference. A party role represents the part played by a party in a given context.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. ENUMERATED with values: * PartyRoleRef

© TM Forum 2026. All Rights Reserved.

Json representation sample(s)

We provide below a JSON representation as example of the 'Document' resource object.


```
{
  "@type": "Document",
  "href": "https://serverRoot/tmf-api/document/v5/document/9555-3c9c417feead",
  "id": "9555-3c9c417feead",
  "name": "Designing Web APIs: Building APIs That Developers Love",
  "documentType": "paperback",
  "version": "1",
  "creationDate": "2021-11-06T10:15:26.007Z",
  "description": "Designing an API is complicated to begin with, but evolving your API
design over time makes the process even more difficult. There are several books on the
topic, but ...",
  "documentSpecification": {
    "@type": "DocumentSpecificationRef",
    "@referredType": "DocumentSpecification",
    "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/a65d-
a08c744f2fe7",
    "id": "a65d-a08c744f2fe7",
    "name": "Standard Paperback Book"
  },
  "lastUpdate": "2024-11-05T10:15:26.007Z",
  "status": "published",
  "attachment": [
    {
      "@type": "Attachment",
      "name": "Link to paperback",
      "url": "https://www.amazon.co.uk/Designing-Web-APIs-Building-
Developers/dp/1492026921",
      "validFor": {
        "startDateTime": "2024-11-05T10:15:26.007Z"
      },
      "attachmentType": "book",
      "mimeType": "text/html"
    }
  ],
  "characteristic": [
    {
      "@type": "IntegerCharacteristic",
      "@baseType": "Characteristic",
      "name": "pages",
      "valueType": "integer",
      "value": 200
    },
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "name": "language",
      "valueType": "string",
      "value": "en"
    },
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "name": "isbn-10",
      "valueType": "string",
```


© TM Forum 2026. All Rights Reserved.

```
      "value": "9781492026921"
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Author",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "http://serverlocation:port/tmf-api/partyManagement/v5/individual/789",
        "id": "789",
        "name": "Brenda Jin",
        "@referredType": "Individual"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Publisher",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "http://serverlocation:port/tmf-
api/partyManagement/v5/organization/123",
        "id": "123",
        "name": "O'Reilly",
        "@referredType": "Organization"
      }
    }
  ]
}
```


DocumentSpecification resource

The template that defines the standard characteristics and expected content for a specific type of document. It serves as a blueprint to ensure all associated document instances share these predefined features. Exemplary values for lifecycleStatus include 'inDesign', 'active', 'rejected', and 'retired'.

Resource model

© TM Forum 2026. All Rights Reserved.

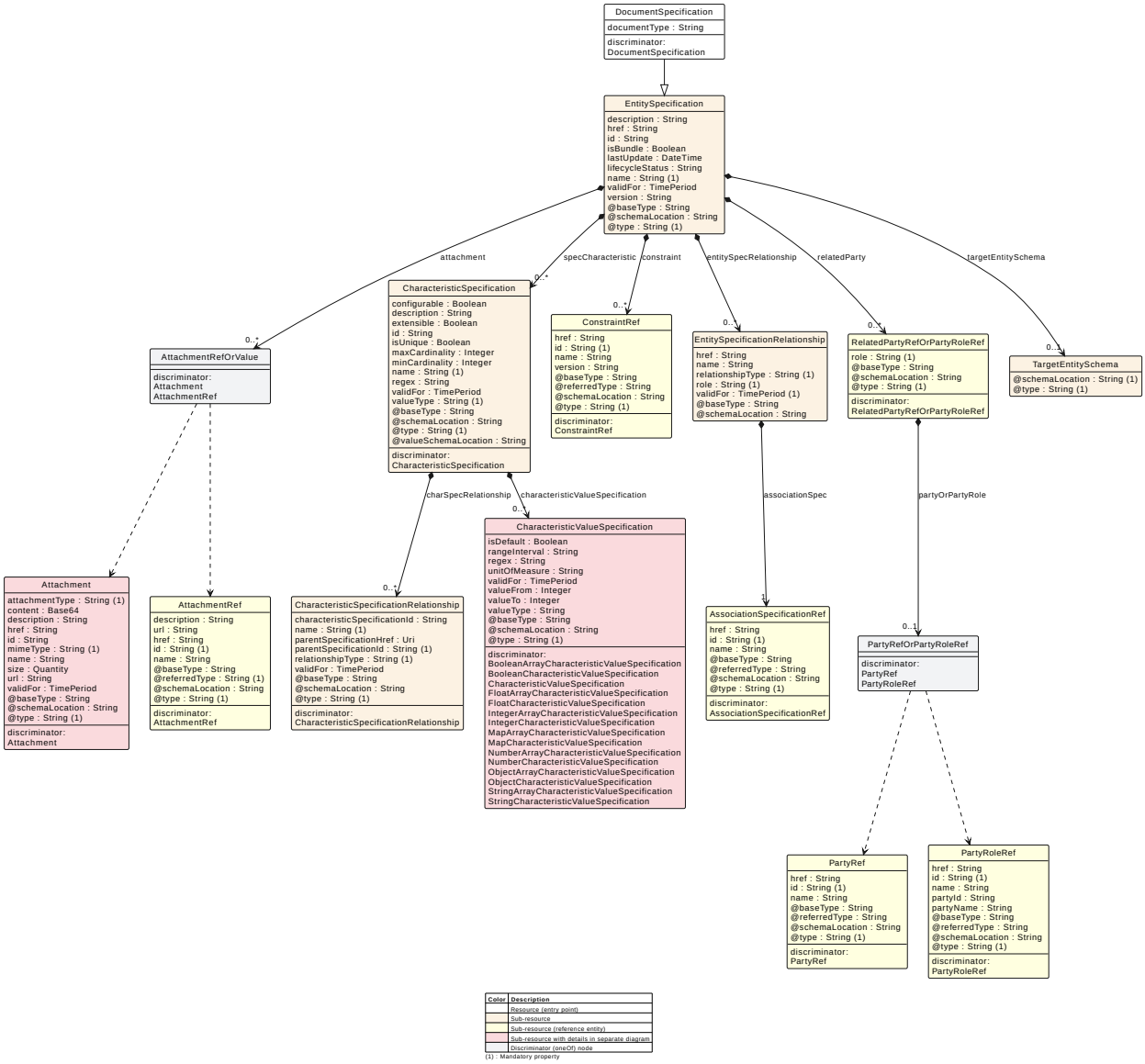


Figure 3 - DocumentSpecification

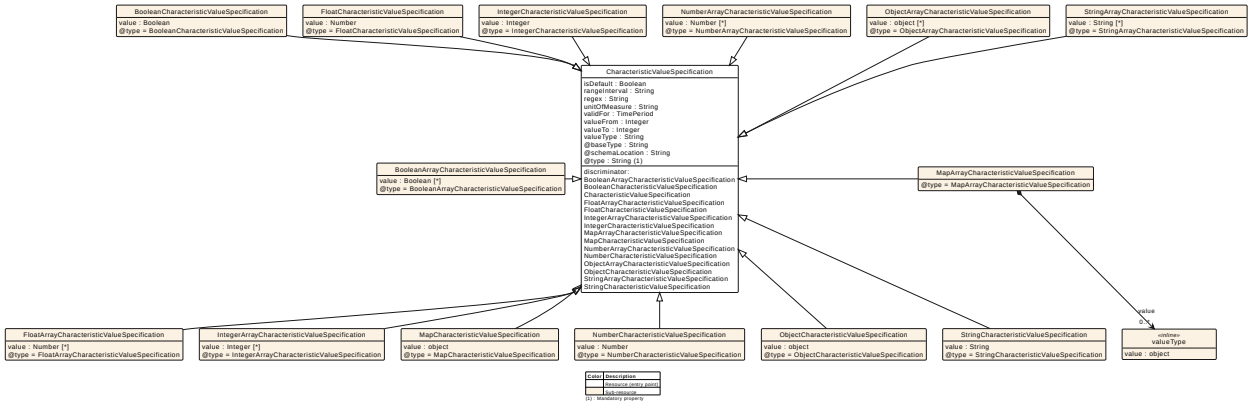


Figure 4 - CharacteristicValueSpecification

© TM Forum 2026. All Rights Reserved.

Field descriptions

DocumentSpecification fields

attachment	An AttachmentRefOrValue [0..*]. Attachments that may be of relevance to this specification, such as picture, document, media.
constraint	A ConstraintRef [0..*]. This is a list of constraint references applied to this specification.
description	A String. Description of the specification.
documentType	A String. The type of document that instances of this specification represent. When a Document is created from this specification, its documentType should be set to this value.
entitySpecRelationship	An EntitySpecificationRelationship [0..*]. Relationship to another specification.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
isBundle	A Boolean. IsBundle determines whether specification represents a single specification (false), or a bundle of specifications (true).
lastUpdate	A DateTime. Date and time of the last update of the specification.
lifecycleStatus	A String. Used to indicate the current lifecycle status of this catalog item.
name	A String [1]. Name given to the specification.
relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. Parties who manage or otherwise have an interest in this specification.
specCharacteristic	A CharacteristicSpecification [0..*]. List of characteristics that the entity can take.
targetEntitySchema	A TargetEntitySchema. The reference object to the schema and type of target entity which is described by a specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. Specification version.

© TM Forum 2026. All Rights Reserved.

@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Attachment sub-resource fields

Complements the description of an element (for instance a product) through video, pictures...

See [Attachment] for details.

AttachmentRef sub-resource fields

Attachment reference. An attachment complements the description of an element (for instance a product) through video, pictures.

See [AttachmentRef] for details.

ConstraintRef sub-resource fields

Constraint reference. The Constraint resource represents a policy/rule applied to an entity or entity spec.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
version	A String. Constraint version.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

EntitySpecificationRelationship sub-resource fields

A migration, substitution, dependency or exclusivity relationship between/among entity

© TM Forum 2026. All Rights Reserved.

specifications.

associationSpec	An AssociationSpecificationRef [1]. Reference to an AssociationSpecification object.
href	A String.
name	A String.
relationshipType	A String [1]. Type of relationship such as migration, substitution, dependency, exclusivity.
role	A String [1]. The association role for this entity specification.
validFor	A TimePeriod [1]. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String.
@schemaLocation	A String.

AssociationSpecificationRef sub-resource fields

Reference to an AssociationSpecification object.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.

See [RelatedPartyRefOrPartyRoleRef] for details.

PartyRef sub-resource fields

© TM Forum 2026. All Rights Reserved.

A Party reference.

See [PartyRef] for details.

PartyRoleRef sub-resource fields

Party role reference. A party role represents the part played by a party in a given context.

See [PartyRoleRef] for details.

CharacteristicSpecification sub-resource fields

This class defines a characteristic specification.

characteristicValueSpecification	A CharacteristicValueSpecification [0..*]. A CharacteristicValueSpecification object is used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a Characteristic Specification object. The values of the attributes in the CharacteristicValueSpecification object describe the values of the attributes that a corresponding Characteristic object can take on. CharacteristicValueSpecification can be instantiated as * BooleanArrayCharacteristicValueSpecification * BooleanCharacteristicValueSpecification * FloatArrayCharacteristicValueSpecification * FloatCharacteristicValueSpecification * IntegerArrayCharacteristicValueSpecification * IntegerCharacteristicValueSpecification * MapArrayCharacteristicValueSpecification * MapCharacteristicValueSpecification * NumberArrayCharacteristicValueSpecification * NumberCharacteristicValueSpecification * ObjectArrayCharacteristicValueSpecification * ObjectCharacteristicValueSpecification * StringArrayCharacteristicValueSpecification * StringCharacteristicValueSpecification
charSpecRelationship	A CharacteristicSpecificationRelationship [0..*]. An aggregation, migration, substitution, dependency or exclusivity relationship between/among Specification Characteristics.

© TM Forum 2026. All Rights Reserved.

configurable	A Boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A String. A narrative that explains the Characteristic Specification.
extensible	A Boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A String. Unique ID for the characteristic.
isUnique	A Boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristic. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An Integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An Integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.
name	A String [1]. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A String. A rule or principle represented in regular expression used to derive the value of a characteristic value.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
valueType	A String [1]. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

© TM Forum 2026. All Rights Reserved.

@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
@valueSchemaLocation	A String. This (optional) field provides a link to the schema describing the value type.

CharacteristicValueSpecification sub-resource fields

Specification of a value (number or text or an object) that can be assigned to a Characteristic. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : BooleanArrayCharacteristicValueSpecification, BooleanCharacteristic ValueSpecification, FloatArrayCharacteristicValueSpecification, FloatCharacteristicValue Specification, IntegerArrayCharacteristicValueSpecification, IntegerCharacteristicValue Specification, MapArrayCharacteristicValueSpecification, MapCharacteristicValue Specification, NumberArrayCharacteristicValueSpecification, NumberCharacteristicValue Specification, ObjectArrayCharacteristicValueSpecification, ObjectCharacteristicValue Specification, StringArrayCharacteristicValueSpecification, StringCharacteristicValue Specification...

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closed Bottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.

© TM Forum 2026. All Rights Reserved.

valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	CharacteristicValueSpecification can be instantiated as * BooleanArrayCharacteristicValueSpecification * BooleanCharacteristicValueSpecification * FloatArrayCharacteristicValueSpecification * FloatCharacteristicValueSpecification * IntegerArrayCharacteristicValueSpecification * IntegerCharacteristicValueSpecification * MapArrayCharacteristicValueSpecification * MapCharacteristicValueSpecification * NumberArrayCharacteristicValueSpecification * NumberCharacteristicValueSpecification * ObjectArrayCharacteristicValueSpecification * ObjectCharacteristicValueSpecification * StringArrayCharacteristicValueSpecification * StringCharacteristicValueSpecification
value	This property is present in subclasses

BooleanArrayCharacteristicValueSpecification sub-resource fields

A specification for a characteristic for which the value is an array booleans.

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closed Bottom" and "closedTop".
regex	A String. A regular expression constraint for given value.

© TM Forum 2026. All Rights Reserved.

unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Boolean [0..*]. Characteristic specification item boolean value.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristicValueSpecification sub-resource fields

A specification for a characteristic for which the value is a boolean.

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closed Bottom" and "closedTop".
regex	A String. A regular expression constraint for given value.

© TM Forum 2026. All Rights Reserved.

unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Boolean. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FloatArrayCharacteristicValueSpecification sub-resource fields

A specification for a characteristic for which the value is a number array of format floating point.

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closed Bottom" and "closedTop".
regex	A String. A regular expression constraint for given value.

© TM Forum 2026. All Rights Reserved.

unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Float [0..*]. Characteristic value.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FloatCharacteristicValueSpecification sub-resource fields

A specification for a characteristic for which the value is a floating point number.

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closed Bottom" and "closedTop".
regex	A String. A regular expression constraint for given value.

© TM Forum 2026. All Rights Reserved.

unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Float. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristicValueSpecification sub-resource fields

A specification for a characteristic for which the value is a number array of format integer.

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closed Bottom" and "closedTop".
regex	A String. A regular expression constraint for given value.

© TM Forum 2026. All Rights Reserved.

unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	An Integer [0..*]. Characteristic value.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristicValueSpecification sub-resource fields

A specification for a characteristic for which the value is an integer number.

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closed Bottom" and "closedTop".
regex	A String. A regular expression constraint for given value.

© TM Forum 2026. All Rights Reserved.

unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	An Integer. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

MapArrayCharacteristicValueSpecification sub-resource fields

A specification for a characteristic for which the value is an array of mapped objects.

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closed Bottom" and "closedTop".
regex	A String. A regular expression constraint for given value.

© TM Forum 2026. All Rights Reserved.

unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A valueType [0..*].
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

valueType sub-resource fields

See [valueType] for details.

valueType_2 sub-resource fields

See [valueType_2] for details.

MapCharacteristicValueSpecification sub-resource fields

A specification for a characteristic for which the value is a mapped object.

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closed Bottom" and "closedTop".

© TM Forum 2026. All Rights Reserved.

regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A valueType_3. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristicValueSpecification sub-resource fields

A specification for a characteristic for which the value is a number array with no specific format.

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closed Bottom" and "closedTop".
regex	A String. A regular expression constraint for given value.

© TM Forum 2026. All Rights Reserved.

unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Number [0..*]. Characteristic value.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristicValueSpecification sub-resource fields

A specification for a characteristic for which the value is a number of no specific format.

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closed Bottom" and "closedTop".
regex	A String. A regular expression constraint for given value.

© TM Forum 2026. All Rights Reserved.

unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Number. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristicValueSpecification sub-resource fields

A specification for a characteristic for which the value is an array of objects.

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closed Bottom" and "closedTop".
regex	A String. A regular expression constraint for given value.

© TM Forum 2026. All Rights Reserved.

unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A valueType_4 [0..*]. Collection of characteristic values.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristicValueSpecification sub-resource fields

A specification for a characteristic for which the value is any object.

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closed Bottom" and "closedTop".
regex	A String. A regular expression constraint for given value.

© TM Forum 2026. All Rights Reserved.

unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A valueType_5. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristicValueSpecification sub-resource fields

A specification for a characteristic for which the value is an array of strings.

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closed Bottom" and "closedTop".
regex	A String. A regular expression constraint for given value.

© TM Forum 2026. All Rights Reserved.

unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A String [0..*]. Characteristic value.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristicValueSpecification sub-resource fields

A specification for a characteristic for which the value is a string.

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closed Bottom" and "closedTop".
regex	A String. A regular expression constraint for given value.

© TM Forum 2026. All Rights Reserved.

unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A String. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

CharacteristicSpecificationRelationship sub-resource fields

An aggregation, migration, substitution, dependency or exclusivity relationship between/ among Characteristic specifications. The specification characteristic is embedded within the specification whose ID and href are in this entity, and identified by its ID.

characteristicSpecificationId	A String. Unique identifier of the characteristic within the specification.
name	A String [1]. Name of the target characteristic within the specification.
parentSpecificationHref	A Uri. Hyperlink reference to the parent specification containing the target characteristic.
parentSpecificationId	A String [1]. Unique identifier of the parent specification containing the target characteristic.
relationshipType	A String [1]. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.

© TM Forum 2026. All Rights Reserved.

validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

TargetEntitySchema sub-resource fields

The reference object to the schema and type of target entity which is described by a specification.

@schemaLocation	A String [1]. This field provides a link to the schema describing the target entity.
@type	A String [1]. Class type of the target entity.

Json representation sample(s)

We provide below a JSON representation as example of the 'DocumentSpecification' resource object.


```
{
  "@type": "DocumentSpecification",
  "@baseType": "EntitySpecification",
  "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/9b12-
789948415bd7",
  "id": "9b12-789948415bd7",
  "name": "Standard Paperback Book",
  "documentType": "paperback",
  "description": "Specification for a Paperback Book",
  "lastUpdate": "2021-11-05T11:03:57.693Z",
  "lifecycleStatus": "published",
  "version": "1.0.0",
  "specCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "pages",
      "valueType": "integer",
      "description": "The number of pages in this paperback book",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "IntegerCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
```


© TM Forum 2026. All Rights Reserved.

```
          "unitOfMeasure": "pages",
          "valueFrom": 1
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "language",
      "valueType": "string",
      "description": "The 2-character ISO 639-1 language code that this book is
published in",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "regex": "^[a-z]{2}"
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "isbn-10",
      "valueType": "string",
      "description": "The 10-digit ISO 2108 International Standard Book Number (ISBN)
of this book",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "regex": "^[0-9]{10}"
        }
      ]
    }
  ]
}
```


Attachment resource

Complements the description of an element (for instance a product) through video, pictures...

Resource model

© TM Forum 2026. All Rights Reserved.

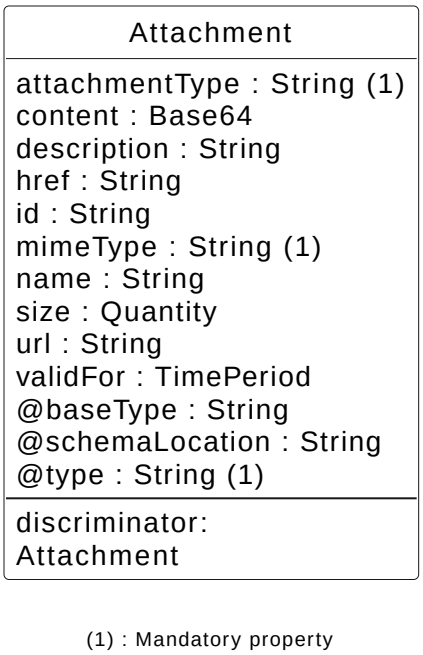


Figure 5 - Attachment

Field descriptions

Attachment fields

attachmentType	A String [1]. A business characterization of the purpose of the attachment, for example logo, instructionManual, contractCopy.
content	A Base64. The actual contents of the attachment object, embedded inline and encoded as base64. Mutually exclusive with url.
description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
mimeType	A String [1]. A technical characterization of the attachment content format using an IETF Media Type, for example application/pdf or image/png. Note: this property is named mimeType for historical reasons.
name	A String. The name of the attachment.
size	A Quantity. An amount in a given unit.

© TM Forum 2026. All Rights Reserved.

url	A String. The URL pointing to the attachment content held in an external system, such as a content management system or document repository. Mutually exclusive with content.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'Attachment' resource object.

```
{
  "@type": "Attachment",
  "href": "https://serverRoot/tmf-api/document/v5/attachment/1492026921",
  "id": "1492026921",
  "name": "Link to paperback",
  "url": "https://www.amazon.co.uk/Designing-Web-APIs-Building-
Developers/dp/1492026921",
  "validFor": {
    "startDateTime": "2024-11-05T10:15:26.007Z"
  },
  "attachmentType": "book",
  "mimeType": "text/html"
}
```

Notification Resource Models

11 notifications are defined for this API.

Notifications related to Document:

- Create Event
- Delete Event
- Attribute Value Change Event
- State Change Event

Notifications related to DocumentSpecification:

© TM Forum 2026. All Rights Reserved.

- Create Event
- Delete Event
- Attribute Value Change Event
- Status Change Event

Notifications related to Attachment:

- Create Event
- Delete Event
- Attribute Value Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

© TM Forum 2026. All Rights Reserved.

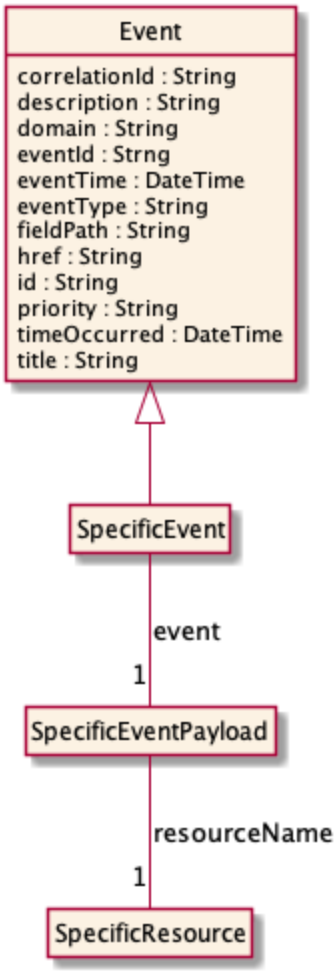


Figure 6 Notification Pattern

Document

Create Event

Message example for DocumentCreateEvent Notification

```
Content-Type: application/json
{
  "@type": "DocumentCreateEvent",
  "eventId": "e3dac91d-a035-465f-afc5-d3b603b53ad2",
  "eventTime": "2021-11-06T10:15:26.007Z",
  "eventType": "DocumentCreateEvent",
  "event": {
    "document": {
      "@type": "Document",
      "href": "https://serverRoot/tmf-api/document/v5/document/9555-3c9c417feead",
      "id": "9555-3c9c417feead",
      "name": "Designing Web APIs: Building APIs That Developers Love",
      "documentType": "paperback",
      "version": "1",
      "creationDate": "2021-11-06T10:15:26.007Z",
```

© TM Forum 2026. All Rights Reserved.

```
      "description": "Designing an API is complicated to begin with, but evolving your
API design over time makes the process even more difficult. There are several books on
the topic, but ...",
      "documentSpecification": {
        "@type": "DocumentSpecificationRef",
        "@referredType": "DocumentSpecification",
        "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/a65d-
a08c744f2fe7",
        "id": "a65d-a08c744f2fe7",
        "name": "Standard Paperback Book"
      },
      "lastUpdate": "2021-11-06T10:15:26.007Z",
      "status": "published",
      "attachment": [
        {
          "@type": "Attachment",
          "name": "Link to paperback",
          "url": "https://www.amazon.co.uk/Designing-Web-APIs-Building-
Developers/dp/1492026921",
          "validFor": {
            "startDateTime": "2021-11-06T10:15:26.007Z"
          },
          "attachmentType": "book",
          "mimeType": "text/html"
        }
      ],
      "characteristic": [
        {
          "@type": "IntegerCharacteristic",
          "@baseType": "Characteristic",
          "name": "pages",
          "valueType": "integer",
          "value": 200
        },
        {
          "@type": "StringCharacteristic",
          "@baseType": "Characteristic",
          "name": "language",
          "valueType": "string",
          "value": "en"
        },
        {
          "@type": "StringCharacteristic",
          "@baseType": "Characteristic",
          "name": "isbn-10",
          "valueType": "string",
          "value": "9781492026921"
        }
      ],
      "relatedParty": [
        {
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "role": "Author",
          "partyOrPartyRole": {
            "@type": "PartyRef",
            "href": "http://serverlocation:port/tmf-
api/partyManagement/v5/individual/789",
            "id": "789",
            "name": "Brenda Jin",
            "@referredType": "Individual"
```


© TM Forum 2026. All Rights Reserved.

```
          }
        },
        {
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "role": "Publisher",
          "partyOrPartyRole": {
            "@type": "PartyRef",
            "href": "http://serverlocation:port/tmf-
api/partyManagement/v5/organization/123",
            "id": "123",
            "name": "O'Reilly",
            "@referredType": "Organization"
          }
        }
      ]
    }
  }
}
```

Delete Event

Message example for DocumentDeleteEvent Notification

```
Content-Type: application/json
{
  "@type": "DocumentDeleteEvent",
  "eventId": "e3dac91d-a035-465f-afc5-d3b603b53ad2",
  "eventTime": "2024-11-10T16:00:00.000Z",
  "eventType": "DocumentDeleteEvent",
  "event": {
    "document": {
      "@type": "Document",
      "href": "https://serverRoot/tmf-api/document/v5/document/9555-3c9c417feead",
      "id": "9555-3c9c417feead"
    }
  }
}
```

Attribute Value Change Event

Message example for DocumentAttributeValueChangeEvent Notification

```
Content-Type: application/json
{
  "@type": "DocumentAttributeValueChangeEvent",
  "eventId": "e3dac91d-a035-465f-afc5-d3b603b53ad2",
  "eventTime": "2024-11-06T09:00:00.000Z",
  "eventType": "DocumentAttributeValueChangeEvent",
  "event": {
    "document": {
      "@type": "Document",
      "href": "https://serverRoot/tmf-api/document/v5/document/9555-3c9c417feead",
      "id": "9555-3c9c417feead",
```

© TM Forum 2026. All Rights Reserved.

```
      "name": "Designing Web APIs: Building APIs That Developers really Love"
    }
  }
}
```

State Change Event

Message example for DocumentStateChangeEvent Notification

```
Content-Type: application/json
{
  "@type": "DocumentStateChangeEvent",
  "eventId": "e3dac91d-a035-465f-afc5-d3b603b53ad2",
  "eventTime": "2024-11-08T14:30:00.000Z",
  "eventType": "DocumentStateChangeEvent",
  "event": {
    "document": {
      "@type": "Document",
      "href": "https://serverRoot/tmf-api/document/v5/document/9555-3c9c417feead",
      "id": "9555-3c9c417feead",
      "status": "published"
    }
  }
}
```

DocumentSpecification

Create Event

Message example for DocumentCreateEvent Notification

```
Content-Type: application/json
{
  "@type": "DocumentSpecificationCreateEvent",
  "eventId": "e3dac91d-a035-465f-afc5-d3b603b53ad2",
  "eventTime": "2021-11-05T11:03:57.693Z",
  "eventType": "DocumentSpecificationCreateEvent",
  "event": {
    "documentSpecification": {
      "@type": "DocumentSpecification",
      "@baseType": "EntitySpecification",
      "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/9b12-
789948415bd7",
      "id": "9b12-789948415bd7",
      "name": "Standard Paperback Book",
      "documentType": "paperback",
      "description": "Specification for a Paperback Book",
      "lastUpdate": "2021-11-05T11:03:57.693Z",
      "lifecycleStatus": "published",
      "version": "1.0.0",
      "specCharacteristic": [
        {
```

© TM Forum 2026. All Rights Reserved.

```
          "@type": "CharacteristicSpecification",
          "name": "pages",
          "valueType": "integer",
          "description": "The number of pages in this paperback book",
          "configurable": true,
          "minCardinality": 1,
          "maxCardinality": 1,
          "extensible": false,
          "characteristicValueSpecification": [
            {
              "@type": "IntegerCharacteristicValueSpecification",
              "@baseType": "CharacteristicValueSpecification",
              "unitOfMeasure": "pages",
              "valueFrom": 1
            }
          ]
        },
        {
          "@type": "CharacteristicSpecification",
          "name": "language",
          "valueType": "string",
          "description": "The 2-character ISO 639-1 language code that this book is
published in",
          "configurable": true,
          "minCardinality": 1,
          "maxCardinality": 1,
          "extensible": false,
          "characteristicValueSpecification": [
            {
              "@type": "StringCharacteristicValueSpecification",
              "@baseType": "CharacteristicValueSpecification",
              "regex": "^[a-z]{2}"
            }
          ]
        },
        {
          "@type": "CharacteristicSpecification",
          "name": "isbn-10",
          "valueType": "string",
          "description": "The 10-digit ISO 2108 International Standard Book Number
(ISBN) of this book",
          "configurable": true,
          "minCardinality": 1,
          "maxCardinality": 1,
          "extensible": false,
          "characteristicValueSpecification": [
            {
              "@type": "StringCharacteristicValueSpecification",
              "@baseType": "CharacteristicValueSpecification",
              "regex": "^[0-9]{10}"
            }
          ]
        }
      ]
    }
  }
}
```

© TM Forum 2026. All Rights Reserved.

Delete Event

Message example for DocumentSpecificationDeleteEvent Notification

```
Content-Type: application/json
{
  "@type": "DocumentSpecificationDeleteEvent",
  "eventId": "e3dac91d-a035-465f-afc5-d3b603b53ad2",
  "eventTime": "2024-11-11T10:00:00.000Z",
  "eventType": "DocumentSpecificationDeleteEvent",
  "event": {
    "documentSpecification": {
      "@type": "DocumentSpecification",
      "@baseType": "EntitySpecification",
      "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/9b12-
789948415bd7",
      "id": "9b12-789948415bd7"
    }
  }
}
```

Attribute Value Change Event

Message example for DocumentSpecificationAttributeValueChangeEvent Notification

```
Content-Type: application/json
{
  "@type": "DocumentSpecificationAttributeValueChangeEvent",
  "eventId": "e3dac91d-a035-465f-afc5-d3b603b53ad2",
  "eventTime": "2021-11-07T09:30:00.000Z",
  "eventType": "DocumentSpecificationAttributeValueChangeEvent",
  "event": {
    "documentSpecification": {
      "@type": "DocumentSpecification",
      "@baseType": "EntitySpecification",
      "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/9b12-
789948415bd7",
      "id": "9b12-789948415bd7",
      "name": "Email"
    }
  }
}
```

Status Change Event

Message example for DocumentSpecificationStatusChangeEvent Notification

```
Content-Type: application/json
{
  "@type": "DocumentSpecificationStatusChangeEvent",
  "eventId": "e3dac91d-a035-465f-afc5-d3b603b53ad2",
```

© TM Forum 2026. All Rights Reserved.

```
  "eventTime": "2021-11-09T14:15:00.000Z",
  "eventType": "DocumentSpecificationStatusChangeEvent",
  "event": {
    "documentSpecification": {
      "@type": "DocumentSpecification",
      "@baseType": "EntitySpecification",
      "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/9b12-
789948415bd7",
      "id": "9b12-789948415bd7",
      "lifecycleStatus": "published"
    }
  }
}
```


Attachment

Create Event

Message example for AttachmentCreateEvent Notification


```
Content-Type: application/json
{
  "@type": "AttachmentCreateEvent",
  "eventId": "e3dac91d-a035-465f-afc5-d3c603b53ad2",
  "eventTime": "2024-11-05T10:15:26.007Z",
  "eventType": "AttachmentCreateEvent",
  "event": {
    "attachment": {
      "@type": "Attachment",
      "href": "https://serverRoot/tmf-api/document/v5/attachment/1492026921",
      "id": "1492026921",
      "name": "Link to paperback",
      "url": "https://www.amazon.co.uk/Designing-Web-APIs-Building-
Developers/dp/1492026921",
      "validFor": {
        "startDateTime": "2024-11-05T10:15:26.007Z"
      },
      "attachmentType": "book",
      "mimeType": "text/html"
    }
  }
}
```


Delete Event

Message example for AttachmentDeleteEvent Notification


```
Content-Type: application/json
{
  "@type": "AttachmentDeleteEvent",
  "eventId": "e3dac91d-a035-465f-afc5-d3c603b53ad2",
  "eventTime": "2024-11-10T16:30:00.000Z",
```


© TM Forum 2026. All Rights Reserved.

```
  "eventType": "AttachmentDeleteEvent",
  "event": {
    "attachment": {
      "@type": "Attachment",
      "href": "https://serverRoot/tmf-api/document/v5/attachment/1492026921",
      "id": "1492026921"
    }
  }
}
```

Attribute Value Change Event

Message example for AttachmentAttributeValueChangeEvent Notification

```
Content-Type: application/json
{
  "@type": "AttachmentAttributeValueChangeEvent",
  "eventId": "e3dac91d-a035-465f-afc5-d3c603b53ad2",
  "eventTime": "2024-11-06T11:00:00.000Z",
  "eventType": "AttachmentAttributeValueChangeEvent",
  "event": {
    "attachment": {
      "@type": "Attachment",
      "href": "https://serverRoot/tmf-api/document/v5/attachment/1492026921",
      "id": "1492026921",
      "name": "Zelda as paperback"
    }
  }
}
```

© TM Forum 2026. All Rights Reserved.

API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on Document

Retrieves a document by ID

`GET /document/{id}?fields=…`

Description

This operation retrieves a Document entity. Attribute selection is enabled for all first level attributes.

Usage samples

A request for a previous Document. The criteria is the Document id 9555-3c9c417feead

Request

© TM Forum 2026. All Rights Reserved.

```
GET /document/9555-3c9c417feead
Content-Type: application/json
```

Response

```
200

{
  "@type": "Document",
  "href": "https://serverRoot/tmf-api/document/v5/document/9555-3c9c417feead",
  "id": "9555-3c9c417feead",
  "name": "Designing Web APIs: Building APIs That Developers Love",
  "documentType": "paperback",
  "version": "1",
  "creationDate": "2021-11-06T10:15:26.007Z",
  "description": "Designing an API is complicated to begin with, but evolving your API
design over time makes the process even more difficult. There are several books on the
topic, but ...",
  "documentSpecification": {
    "@type": "DocumentSpecificationRef",
    "@referredType": "DocumentSpecification",
    "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/a65d-
a08c744f2fe7",
    "id": "a65d-a08c744f2fe7",
    "name": "Standard Paperback Book"
  },
  "lastUpdate": "2021-11-06T10:15:26.007Z",
  "status": "published",
  "attachment": [
    {
      "@type": "AttachmentRef",
      "@referredType": "Attachment",
      "href": "https://serverRoot/tmf-api/document/v5/attachment/1492026921",
      "id": "1492026921",
      "name": "Link to paperback"
    }
  ],
  "characteristic": [
    {
      "@type": "IntegerCharacteristic",
      "@baseType": "Characteristic",
      "name": "pages",
      "valueType": "integer",
      "value": 200
    },
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "name": "language",
      "valueType": "string",
      "value": "en"
    },
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "name": "isbn-10",
```

© TM Forum 2026. All Rights Reserved.

```
      "valueType": "string",
      "value": "9781492026921"
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Author",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "http://serverlocation:port/tmf-api/partyManagement/v5/individual/789",
        "id": "789",
        "name": "Brenda Jin",
        "@referredType": "Individual"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Publisher",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "http://serverlocation:port/tmf-
api/partyManagement/v5/organization/123",
        "id": "123",
        "name": "O'Reilly",
        "@referredType": "Organization"
      }
    }
  ]
}
```


List or find document objects

`GET /document?fields=…&{filtering}`

Description

This operation list Document entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieving previous Document details for the documentType of 'paperback'

Request

```
GET /document?filter=documentType=paperback
Content-Type: application/json
```

Response

© TM Forum 2026. All Rights Reserved.

```
200

[
  {
    "@type": "Document",
    "href": "https://serverRoot/tmf-api/document/v5/document/9555-3c9c417feead",
    "id": "9555-3c9c417feead",
    "name": "Designing Web APIs: Building APIs That Developers Love",
    "documentType": "paperback",
    "version": "1",
    "creationDate": "2021-11-06T10:15:26.007Z",
    "description": "Designing an API is complicated to begin with, but evolving your
API design over time makes the process even more difficult. There are several books on
the topic, but ...",
    "documentSpecification": {
      "@type": "DocumentSpecificationRef",
      "@referredType": "DocumentSpecification",
      "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/a65d-
a08c744f2fe7",
      "id": "a65d-a08c744f2fe7",
      "name": "Standard Paperback Book"
    },
    "lastUpdate": "2024-11-05T10:15:26.007Z",
    "status": "published",
    "attachment": [
      {
        "@type": "Attachment",
        "name": "Link to paperback",
        "url": "https://www.amazon.co.uk/Designing-Web-APIs-Building-
Developers/dp/1492026921",
        "validFor": {
          "startDateTime": "2024-11-05T10:15:26.007Z"
        },
        "attachmentType": "book",
        "mimeType": "text/html"
      }
    ],
    "characteristic": [
      {
        "@type": "IntegerCharacteristic",
        "@baseType": "Characteristic",
        "name": "pages",
        "valueType": "integer",
        "value": 200
      },
      {
        "@type": "StringCharacteristic",
        "@baseType": "Characteristic",
        "name": "language",
        "valueType": "string",
        "value": "en"
      },
      {
        "@type": "StringCharacteristic",
        "@baseType": "Characteristic",
        "name": "isbn-10",
        "valueType": "string",
        "value": "9781492026921"
      }
```


© TM Forum 2026. All Rights Reserved.

```
    ],
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "Author",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "http://serverlocation:port/tmf-
api/partyManagement/v5/individual/789",
          "id": "789",
          "name": "Brenda Jin",
          "@referredType": "Individual"
        }
      },
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "Publisher",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "http://serverlocation:port/tmf-
api/partyManagement/v5/organization/123",
          "id": "123",
          "name": "O'Reilly",
          "@referredType": "Organization"
        }
      }
    ]
  }
]
```

Creates a document

`POST /document?fields=…&{filtering}`

Description

This operation creates a Document entity.

Mandatory Attributes

Mandatory Attributes	Rule
name	
@type	

Usage samples

Creating a Document

Request

```
POST /document
```

© TM Forum 2026. All Rights Reserved.

```
Content-Type: application/json

{
  "@type": "Document",
  "name": "Designing Web APIs: Building APIs That Developers Love",
  "documentType": "paperback",
  "version": "1",
  "description": "Designing an API is complicated to begin with, but evolving your API
design over time makes the process even more difficult. There are several books on the
topic, but ...",
  "documentSpecification": {
    "@type": "DocumentSpecificationRef",
    "@referredType": "DocumentSpecification",
    "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/a65d-
a08c744f2fe7",
    "id": "a65d-a08c744f2fe7",
    "name": "Standard Paperback Book"
  },
  "status": "published",
  "attachment": [
    {
      "@type": "AttachmentRef",
      "@referredType": "Attachment",
      "href": "https://serverRoot/tmf-api/document/v5/attachment/1492026921",
      "id": "1492026921",
      "name": "Link to paperback"
    }
  ],
  "characteristic": [
    {
      "@type": "IntegerCharacteristic",
      "@baseType": "Characteristic",
      "name": "pages",
      "valueType": "integer",
      "value": 200
    },
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "name": "language",
      "valueType": "string",
      "value": "en"
    },
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "name": "isbn-10",
      "valueType": "string",
      "value": "9781492026921"
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Author",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "http://serverlocation:port/tmf-api/partyManagement/v5/individual/789",
        "id": "789",
        "name": "Brenda Jin",
```

© TM Forum 2026. All Rights Reserved.

```
        "@referredType": "Individual"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Publisher",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "http://serverlocation:port/tmf-
api/partyManagement/v5/organization/123",
        "id": "123",
        "name": "O'Reilly",
        "@referredType": "Organization"
      }
    }
  ]
}
```

Response

```
201

{
  "@type": "Document",
  "href": "https://serverRoot/tmf-api/document/v5/document/9555-3c9c417feead",
  "id": "9555-3c9c417feead",
  "name": "Designing Web APIs: Building APIs That Developers Love",
  "documentType": "paperback",
  "version": "1",
  "creationDate": "2021-11-06T10:15:26.007Z",
  "description": "Designing an API is complicated to begin with, but evolving your API
design over time makes the process even more difficult. There are several books on the
topic, but ...",
  "documentSpecification": {
    "@type": "DocumentSpecificationRef",
    "@referredType": "DocumentSpecification",
    "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/a65d-
a08c744f2fe7",
    "id": "a65d-a08c744f2fe7",
    "name": "Standard Paperback Book"
  },
  "lastUpdate": "2021-11-06T10:15:26.007Z",
  "status": "published",
  "attachment": [
    {
      "@type": "AttachmentRef",
      "@referredType": "Attachment",
      "href": "https://serverRoot/tmf-api/document/v5/attachment/1492026921",
      "id": "1492026921",
      "name": "Link to paperback"
    }
  ],
  "characteristic": [
    {
      "@type": "IntegerCharacteristic",
      "@baseType": "Characteristic",
      "name": "pages",
```

© TM Forum 2026. All Rights Reserved.

```
      "valueType": "integer",
      "value": 200
    },
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "name": "language",
      "valueType": "string",
      "value": "en"
    },
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "name": "isbn-10",
      "valueType": "string",
      "value": "9781492026921"
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Author",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "http://serverlocation:port/tmf-api/partyManagement/v5/individual/789",
        "id": "789",
        "name": "Brenda Jin",
        "@referredType": "Individual"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Publisher",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "http://serverlocation:port/tmf-
api/partyManagement/v5/organization/123",
        "id": "123",
        "name": "O'Reilly",
        "@referredType": "Organization"
      }
    }
  ]
}
```


Updates partially a document

PATCH /document/{id}?fields=…

Description

This operation allows partial updates of a Document entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-

© TM Forum 2026. All Rights Reserved.

resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
creationDate	
href	
id	
lastUpdate	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
attachment	
category	
characteristic	
description	
documentRelationship	
documentSpecification	
documentType	
expiryDate	
externalIdentifier	
name	
relatedEntity	
relatedParty	
status	
validFor	
version	

Usage samples

Here is an example of a request for patching an document.

© TM Forum 2026. All Rights Reserved.

Request

```
PATCH /document/9555-3c9c417feead
Content-Type: application/json

{
  "@type": "Document",
  "name": "Designing Web APIs: Building APIs That Developers really Love"
}
```

Response

```
200

{
  "@type": "Document",
  "href": "https://serverRoot/tmf-api/document/v5/document/9555-3c9c417feead",
  "id": "9555-3c9c417feead",
  "name": "Designing Web APIs: Building APIs That Developers really Love",
  "documentType": "paperback",
  "version": "1",
  "creationDate": "2021-11-06T10:15:26.007Z",
  "description": "Designing an API is complicated to begin with, but evolving your API
design over time makes the process even more difficult. There are several books on the
topic, but ...",
  "documentSpecification": {
    "@type": "DocumentSpecificationRef",
    "@referredType": "DocumentSpecification",
    "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/a65d-
a08c744f2fe7",
    "id": "a65d-a08c744f2fe7",
    "name": "Standard Paperback Book"
  },
  "lastUpdate": "2024-11-05T10:15:26.007Z",
  "status": "published",
  "attachment": [
    {
      "@type": "Attachment",
      "name": "Link to paperback",
      "url": "https://www.amazon.co.uk/Designing-Web-APIs-Building-
Developers/dp/1492026921",
      "validFor": {
        "startDateTime": "2024-11-05T10:15:26.007Z"
      },
      "attachmentType": "book",
      "mimeType": "text/html"
    }
  ],
  "characteristic": [
    {
      "@type": "IntegerCharacteristic",
      "@baseType": "Characteristic",
      "name": "pages",
      "valueType": "integer",
      "value": 200
    },
```

© TM Forum 2026. All Rights Reserved.

```
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "name": "language",
      "valueType": "string",
      "value": "en"
    },
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "name": "isbn-10",
      "valueType": "string",
      "value": "9781492026921"
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Author",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "http://serverlocation:port/tmf-api/partyManagement/v5/individual/789",
        "id": "789",
        "name": "Brenda Jin",
        "@referredType": "Individual"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Publisher",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "http://serverlocation:port/tmf-
api/partyManagement/v5/organization/123",
        "id": "123",
        "name": "O'Reilly",
        "@referredType": "Organization"
      }
    }
  ]
}
```


Here is an example of a request for patching an document.

Request

```
PATCH /document/9555-3c9c417feead
Content-Type: application/merge-patch+json

{
  "@type": "Document",
  "name": "Designing Web APIs: Building APIs That Developers really Love"
}
```

Response

© TM Forum 2026. All Rights Reserved.

```
200

{
  "@type": "Document",
  "href": "https://serverRoot/tmf-api/document/v5/document/9555-3c9c417feead",
  "id": "9555-3c9c417feead",
  "name": "Designing Web APIs: Building APIs That Developers really Love",
  "documentType": "paperback",
  "version": "1",
  "creationDate": "2021-11-06T10:15:26.007Z",
  "description": "Designing an API is complicated to begin with, but evolving your API
design over time makes the process even more difficult. There are several books on the
topic, but ...",
  "documentSpecification": {
    "@type": "DocumentSpecificationRef",
    "@referredType": "DocumentSpecification",
    "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/a65d-
a08c744f2fe7",
    "id": "a65d-a08c744f2fe7",
    "name": "Standard Paperback Book"
  },
  "lastUpdate": "2024-11-05T10:15:26.007Z",
  "status": "published",
  "attachment": [
    {
      "@type": "Attachment",
      "name": "Link to paperback",
      "url": "https://www.amazon.co.uk/Designing-Web-APIs-Building-
Developers/dp/1492026921",
      "validFor": {
        "startDateTime": "2024-11-05T10:15:26.007Z"
      },
      "attachmentType": "book",
      "mimeType": "text/html"
    }
  ],
  "characteristic": [
    {
      "@type": "IntegerCharacteristic",
      "@baseType": "Characteristic",
      "name": "pages",
      "valueType": "integer",
      "value": 200
    },
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "name": "language",
      "valueType": "string",
      "value": "en"
    },
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "name": "isbn-10",
      "valueType": "string",
      "value": "9781492026921"
    }
  ],
```

© TM Forum 2026. All Rights Reserved.

```
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Author",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "http://serverlocation:port/tmf-api/partyManagement/v5/individual/789",
        "id": "789",
        "name": "Brenda Jin",
        "@referredType": "Individual"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Publisher",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "http://serverlocation:port/tmf-
api/partyManagement/v5/organization/123",
        "id": "123",
        "name": "O'Reilly",
        "@referredType": "Organization"
      }
    }
  ]
}
```

Here is an example of a request for patching an document.

Request

```
PATCH /document/9555-3c9c417feead
Content-Type: application/json-patch+json

[
  {
    "op": "replace",
    "path": "/name",
    "value": "Designing Web APIs: Building APIs That Developers really Love"
  },
  {
    "op": "replace",
    "path": "/version",
    "value": "2"
  }
]
```

Response

```
200

{
  "@type": "Document",
  "href": "https://serverRoot/tmf-api/document/v5/document/9555-3c9c417feead",
```

© TM Forum 2026. All Rights Reserved.

```
  "id": "9555-3c9c417feead",
  "name": "Designing Web APIs: Building APIs That Developers really Love",
  "documentType": "paperback",
  "version": "2",
  "creationDate": "2021-11-06T10:15:26.007Z",
  "description": "Designing an API is complicated to begin with, but evolving your API
design over time makes the process even more difficult. There are several books on the
topic, but ...",
  "documentSpecification": {
    "@type": "DocumentSpecificationRef",
    "@referredType": "DocumentSpecification",
    "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/a65d-
a08c744f2fe7",
    "id": "a65d-a08c744f2fe7",
    "name": "Standard Paperback Book"
  },
  "lastUpdate": "2024-11-05T10:15:26.007Z",
  "status": "published",
  "attachment": [
    {
      "@type": "Attachment",
      "name": "Link to paperback",
      "url": "https://www.amazon.co.uk/Designing-Web-APIs-Building-
Developers/dp/1492026921",
      "validFor": {
        "startDateTime": "2024-11-05T10:15:26.007Z"
      },
      "attachmentType": "book",
      "mimeType": "text/html"
    }
  ],
  "characteristic": [
    {
      "@type": "IntegerCharacteristic",
      "@baseType": "Characteristic",
      "name": "pages",
      "valueType": "integer",
      "value": 200
    },
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "name": "language",
      "valueType": "string",
      "value": "en"
    },
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "name": "isbn-10",
      "valueType": "string",
      "value": "9781492026921"
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Author",
      "partyOrPartyRole": {
        "@type": "PartyRef",
```


© TM Forum 2026. All Rights Reserved.

```
        "href": "http://serverlocation:port/tmf-api/partyManagement/v5/individual/789",
        "id": "789",
        "name": "Brenda Jin",
        "@referredType": "Individual"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Publisher",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "http://serverlocation:port/tmf-
api/partyManagement/v5/organization/123",
        "id": "123",
        "name": "O'Reilly",
        "@referredType": "Organization"
      }
    }
  ]
}
```

Here is an example of a request for patching an document.

Request

```
PATCH /document/9555-3c9c417feead
Content-Type: application/json-patch-query+json

[
  {
    "op": "replace",
    "path": "/name",
    "value": "Designing Web APIs: Building APIs That Developers really Love"
  },
  {
    "op": "replace",
    "path": "/characteristic?name=pages",
    "value": {
      "@type": "IntegerCharacteristic",
      "@baseType": "Characteristic",
      "name": "pages",
      "valueType": "integer",
      "value": 212
    }
  }
]
```

Response

```
200

{
  "@type": "Document",
  "href": "https://serverRoot/tmf-api/document/v5/document/9555-3c9c417feead",
```

© TM Forum 2026. All Rights Reserved.

```
  "id": "9555-3c9c417feead",
  "name": "Designing Web APIs: Building APIs That Developers really Love",
  "documentType": "paperback",
  "version": "1",
  "creationDate": "2021-11-06T10:15:26.007Z",
  "description": "Designing an API is complicated to begin with, but evolving your API
design over time makes the process even more difficult. There are several books on the
topic, but ...",
  "documentSpecification": {
    "@type": "DocumentSpecificationRef",
    "@referredType": "DocumentSpecification",
    "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/a65d-
a08c744f2fe7",
    "id": "a65d-a08c744f2fe7",
    "name": "Standard Paperback Book"
  },
  "lastUpdate": "2024-11-05T10:15:26.007Z",
  "status": "published",
  "attachment": [
    {
      "@type": "Attachment",
      "name": "Link to paperback",
      "url": "https://www.amazon.co.uk/Designing-Web-APIs-Building-
Developers/dp/1492026921",
      "validFor": {
        "startDateTime": "2024-11-05T10:15:26.007Z"
      },
      "attachmentType": "book",
      "mimeType": "text/html"
    }
  ],
  "characteristic": [
    {
      "@type": "IntegerCharacteristic",
      "@baseType": "Characteristic",
      "name": "pages",
      "valueType": "integer",
      "value": 212
    },
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "name": "language",
      "valueType": "string",
      "value": "en"
    },
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "name": "isbn-10",
      "valueType": "string",
      "value": "9781492026921"
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Author",
      "partyOrPartyRole": {
        "@type": "PartyRef",
```


© TM Forum 2026. All Rights Reserved.

```
        "href": "http://serverlocation:port/tmf-api/partyManagement/v5/individual/789",
        "id": "789",
        "name": "Brenda Jin",
        "@referredType": "Individual"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Publisher",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "http://serverlocation:port/tmf-
api/partyManagement/v5/organization/123",
        "id": "123",
        "name": "O'Reilly",
        "@referredType": "Organization"
      }
    }
  ]
}
```


Deletes a document

`DELETE /document/{id}?{filtering}`

Description

This operation deletes a Document entity.

Operations on DocumentSpecification

Retrieves a documentSpecification by ID

`GET /documentSpecification/{id}?fields=…`

Description

This operation retrieves a DocumentSpecification entity. Attribute selection is enabled for all first level attributes.

Usage samples

A request for a specific DocumentSpecification with an id 9b12-789948415bd7

Request

```
GET /documentSpecification/9b12-789948415bd7
Content-Type: application/json
```

© TM Forum 2026. All Rights Reserved.

Response

```
200

{
  "@type": "DocumentSpecification",
  "@baseType": "EntitySpecification",
  "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/9b12-
789948415bd7",
  "id": "9b12-789948415bd7",
  "name": "Standard Paperback Book",
  "documentType": "paperback",
  "description": "Specification for a Paperback Book",
  "lastUpdate": "2021-11-05T11:03:57.693Z",
  "lifecycleStatus": "published",
  "version": "1.0.0",
  "specCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "pages",
      "valueType": "integer",
      "description": "The number of pages in this paperback book",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "IntegerCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "unitOfMeasure": "pages",
          "valueFrom": 1
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "language",
      "valueType": "string",
      "description": "The 2-character ISO 639-1 language code that this book is
published in",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "regex": "^[a-z]{2}"
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "isbn-10",
      "valueType": "string",
      "description": "The 10-digit ISO 2108 International Standard Book Number (ISBN)
```

© TM Forum 2026. All Rights Reserved.

```
of this book",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "regex": "^[0-9]{10}"
        }
      ]
    }
  ]
}
```

List or find documentSpecification objects

`GET /documentSpecification?fields=…&{filtering}`

Description

This operation list DocumentSpecification entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieving details of DocumentSpecifications which are published

Request

```
GET /documentSpecification?filter=status=published
Content-Type: application/json
```

Response

```
200

[
  {
    "@type": "DocumentSpecification",
    "@baseType": "EntitySpecification",
    "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/9b12-
789948415bd7",
    "id": "9b12-789948415bd7",
    "name": "Standard Paperback Book",
    "documentType": "paperback",
    "description": "Specification for a Paperback Book",
    "lastUpdate": "2021-11-05T11:03:57.693Z",
    "lifecycleStatus": "published",
```

© TM Forum 2026. All Rights Reserved.

```
    "version": "1.0.0",
    "specCharacteristic": [
      {
        "@type": "CharacteristicSpecification",
        "name": "pages",
        "valueType": "integer",
        "description": "The number of pages in this paperback book",
        "configurable": true,
        "minCardinality": 1,
        "maxCardinality": 1,
        "extensible": false,
        "characteristicValueSpecification": [
          {
            "@type": "IntegerCharacteristicValueSpecification",
            "@baseType": "CharacteristicValueSpecification",
            "unitOfMeasure": "pages",
            "valueFrom": 1
          }
        ]
      },
      {
        "@type": "CharacteristicSpecification",
        "name": "language",
        "valueType": "string",
        "description": "The 2-character ISO 639-1 language code that this book is
published in",
        "configurable": true,
        "minCardinality": 1,
        "maxCardinality": 1,
        "extensible": false,
        "characteristicValueSpecification": [
          {
            "@type": "StringCharacteristicValueSpecification",
            "@baseType": "CharacteristicValueSpecification",
            "regex": "^[a-z]{2}"
          }
        ]
      },
      {
        "@type": "CharacteristicSpecification",
        "name": "isbn-10",
        "valueType": "string",
        "description": "The 10-digit ISO 2108 International Standard Book Number (ISBN)
of this book",
        "configurable": true,
        "minCardinality": 1,
        "maxCardinality": 1,
        "extensible": false,
        "characteristicValueSpecification": [
          {
            "@type": "StringCharacteristicValueSpecification",
            "@baseType": "CharacteristicValueSpecification",
            "regex": "^[0-9]{10}"
          }
        ]
      }
    ]
  }
]
```


© TM Forum 2026. All Rights Reserved.

Creates a documentSpecification

`POST /documentSpecification?fields=…&{filtering}`

Description

This operation creates a DocumentSpecification entity.

Mandatory Attributes

Mandatory Attributes	Rule
@type	

Usage samples

Creating a DocumentSpecification

Request

```
POST /documentSpecification
Content-Type: application/json

{
  "@type": "DocumentSpecification",
  "@baseType": "EntitySpecification",
  "name": "Standard Paperback Book",
  "documentType": "paperback",
  "description": "Specification for a Paperback Book",
  "lifecycleStatus": "published",
  "version": "1.0.0",
  "specCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "pages",
      "valueType": "integer",
      "description": "The number of pages in this paperback book",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "IntegerCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "unitOfMeasure": "pages",
          "valueFrom": 1
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "language",
      "valueType": "string",
```

© TM Forum 2026. All Rights Reserved.

```
      "description": "The 2-character ISO 639-1 language code that this book is
published in",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "regex": "^[a-z]{2}"
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "isbn-10",
      "valueType": "string",
      "description": "The 10-digit ISO 2108 International Standard Book Number (ISBN)
of this book",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "regex": "^[0-9]{10}"
        }
      ]
    }
  ]
}
```

Response

```
201

{
  "@type": "DocumentSpecification",
  "@baseType": "EntitySpecification",
  "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/9b12-
789948415bd7",
  "id": "9b12-789948415bd7",
  "name": "Standard Paperback Book",
  "documentType": "paperback",
  "description": "Specification for a Paperback Book",
  "lastUpdate": "2021-11-05T11:03:57.693Z",
  "lifecycleStatus": "published",
  "version": "1.0.0",
  "specCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "pages",
      "valueType": "integer",
      "description": "The number of pages in this paperback book",
```

© TM Forum 2026. All Rights Reserved.

```
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "IntegerCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "unitOfMeasure": "pages",
          "valueFrom": 1
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "language",
      "valueType": "string",
      "description": "The 2-character ISO 639-1 language code that this book is
published in",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "regex": "^[a-z]{2}"
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "isbn-10",
      "valueType": "string",
      "description": "The 10-digit ISO 2108 International Standard Book Number (ISBN)
of this book",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "regex": "^[0-9]{10}"
        }
      ]
    }
  ]
}
```


Updates partially a documentSpecification

`PATCH /documentSpecification/{id}?fields=…`

Description

© TM Forum 2026. All Rights Reserved.

This operation allows partial updates of a DocumentSpecification entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
href	
id	
lastUpdate	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
attachment	
constraint	
description	
documentType	
entitySpecRelationship	
isBundle	
lifecycleStatus	
name	
relatedParty	
specCharacteristic	
targetEntitySchema	
validFor	
version	

Usage samples

Here is an example of a request for patching an document specification.

© TM Forum 2026. All Rights Reserved.

Request

```
PATCH /documentSpecification/9b12-789948415bd7
Content-Type: application/json

{
  "@type": "DocumentSpecification",
  "@baseType": "EntitySpecification",
  "name": "Email"
}
```

Response

```
200

{
  "@type": "DocumentSpecification",
  "@baseType": "EntitySpecification",
  "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/9b12-
789948415bd7",
  "id": "9b12-789948415bd7",
  "name": "Email",
  "documentType": "paperback",
  "description": "Specification for a Paperback Book",
  "lastUpdate": "2021-11-05T11:03:57.693Z",
  "lifecycleStatus": "published",
  "version": "1.0.0",
  "specCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "pages",
      "valueType": "integer",
      "description": "The number of pages in this paperback book",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "IntegerCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "unitOfMeasure": "pages",
          "valueFrom": 1
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "language",
      "valueType": "string",
      "description": "The 2-character ISO 639-1 language code that this book is
published in",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
```

© TM Forum 2026. All Rights Reserved.

```
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "regex": "^[a-z]{2}"
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "isbn-10",
      "valueType": "string",
      "description": "The 10-digit ISO 2108 International Standard Book Number (ISBN)
of this book",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "regex": "^[0-9]{10}"
        }
      ]
    }
  ]
}
```

Here is an example of a request for patching an document specification .

Request

```
PATCH /documentSpecification/9b12-789948415bd7
Content-Type: application/merge-patch+json

{
  "@type": "DocumentSpecification",
  "@baseType": "EntitySpecification",
  "name": "Email"
}
```

Response

```
200

{
  "@type": "DocumentSpecification",
  "@baseType": "EntitySpecification",
  "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/9b12-
789948415bd7",
  "id": "9b12-789948415bd7",
  "name": "Email",
  "documentType": "paperback",
```

© TM Forum 2026. All Rights Reserved.

```
  "description": "Specification for a Paperback Book",
  "lastUpdate": "2021-11-05T11:03:57.693Z",
  "lifecycleStatus": "published",
  "version": "1.0.0",
  "specCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "pages",
      "valueType": "integer",
      "description": "The number of pages in this paperback book",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "IntegerCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "unitOfMeasure": "pages",
          "valueFrom": 1
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "language",
      "valueType": "string",
      "description": "The 2-character ISO 639-1 language code that this book is
published in",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "regex": "^[a-z]{2}"
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "isbn-10",
      "valueType": "string",
      "description": "The 10-digit ISO 2108 International Standard Book Number (ISBN)
of this book",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "regex": "^[0-9]{10}"
        }
      ]
    }
  ]
```


© TM Forum 2026. All Rights Reserved.

```
}
```

Here is an example of a request for patching an document.

Request

```
PATCH /documentSpecification/9b12-789948415bd7
Content-Type: application/json-patch+json

[
  {
    "op": "replace",
    "path": "/name",
    "value": "Email"
  },
  {
    "op": "replace",
    "path": "/version",
    "value": "1.4.5"
  }
]
```

Response

```
200

{
  "@type": "DocumentSpecification",
  "@baseType": "EntitySpecification",
  "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/9b12-
789948415bd7",
  "id": "9b12-789948415bd7",
  "name": "Email",
  "documentType": "paperback",
  "description": "Specification for a Paperback Book",
  "lastUpdate": "2021-11-05T11:03:57.693Z",
  "lifecycleStatus": "published",
  "version": "1.4.5",
  "specCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "pages",
      "valueType": "integer",
      "description": "The number of pages in this paperback book",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "IntegerCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "unitOfMeasure": "pages",
          "valueFrom": 1
```

© TM Forum 2026. All Rights Reserved.

```
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "language",
      "valueType": "string",
      "description": "The 2-character ISO 639-1 language code that this book is
published in",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "regex": "^[a-z]{2}"
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "isbn-10",
      "valueType": "string",
      "description": "The 10-digit ISO 2108 International Standard Book Number (ISBN)
of this book",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "regex": "^[0-9]{10}"
        }
      ]
    }
  ]
}
```

Here is an example of a request for patching an document.

Request

```
PATCH /documentSpecification/9b12-789948415bd7
Content-Type: application/json-patch-query+json

[
  {
    "op": "replace",
    "path": "/name",
    "value": "Email"
  },
  {
    "op": "replace",
```

© TM Forum 2026. All Rights Reserved.

```
    "path": "/specCharacteristic?name=pages",
    "value": {
      "@type": "CharacteristicSpecification",
      "name": "pages",
      "valueType": "integer",
      "description": "The number of pages in this paperback book",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "IntegerCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "unitOfMeasure": "pages",
          "valueFrom": 10
        }
      ]
    }
  }
]
```

Response

```
200

{
  "@type": "DocumentSpecification",
  "@baseType": "EntitySpecification",
  "href": "https://serverRoot/tmf-api/document/v5/documentSpecification/9b12-
789948415bd7",
  "id": "9b12-789948415bd7",
  "name": "Email",
  "documentType": "paperback",
  "description": "Specification for a Paperback Book",
  "lastUpdate": "2021-11-05T11:03:57.693Z",
  "lifecycleStatus": "published",
  "version": "1.0.0",
  "specCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "pages",
      "valueType": "integer",
      "description": "The number of pages in this paperback book",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "IntegerCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "unitOfMeasure": "pages",
          "valueFrom": 10
        }
      ]
    },
```

© TM Forum 2026. All Rights Reserved.

```
    {
      "@type": "CharacteristicSpecification",
      "name": "language",
      "valueType": "string",
      "description": "The 2-character ISO 639-1 language code that this book is
published in",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "regex": "^[a-z]{2}"
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "name": "isbn-10",
      "valueType": "string",
      "description": "The 10-digit ISO 2108 International Standard Book Number (ISBN)
of this book",
      "configurable": true,
      "minCardinality": 1,
      "maxCardinality": 1,
      "extensible": false,
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "@baseType": "CharacteristicValueSpecification",
          "regex": "^[0-9]{10}"
        }
      ]
    }
  ]
}
```


Deletes a documentSpecification

`DELETE /documentSpecification/{id}?{filtering}`

Description

This operation deletes a DocumentSpecification entity.

Operations on Attachment

Retrieves a attachment by ID

`GET /attachment/{id}?fields=…`

Description

© TM Forum 2026. All Rights Reserved.

This operation retrieves an Attachment entity. Attribute selection is enabled for all first level attributes.

Usage samples

A request for a previous Attachment. The criteria is the Attachment id 1492026921

Request

```
GET /attachment/1492026921
Content-Type: application/json
```

Response

```
200

{
  "@type": "Attachment",
  "href": "https://serverRoot/tmf-api/document/v5/attachment/1492026921",
  "id": "1492026921",
  "name": "Link to paperback",
  "url": "https://www.amazon.co.uk/Designing-Web-APIs-Building-
Developers/dp/1492026921",
  "validFor": {
    "startDateTime": "2024-11-05T10:15:26.007Z"
  },
  "attachmentType": "book",
  "mimeType": "text/html"
}
```

List or find attachment objects

`GET /attachment?fields=…&{filtering}`

Description

This operation list Attachment entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieving previous Attachment details for the attachmentType of 'book'

Request

```
GET /attachment?filter=attachmentType=book
```

© TM Forum 2026. All Rights Reserved.

```
Content-Type: application/json
```

Response

```
200

[
  {
    "@type": "Attachment",
    "href": "https://serverRoot/tmf-api/document/v5/attachment/1492026921",
    "id": "1492026921",
    "name": "Link to paperback",
    "url": "https://www.amazon.co.uk/Designing-Web-APIs-Building-
Developers/dp/1492026921",
    "validFor": {
      "startDateTime": "2024-11-05T10:15:26.007Z"
    },
    "attachmentType": "book",
    "mimeType": "text/html"
  }
]
```

Creates a attachment

`POST /attachment?fields=…&{filtering}`

Description

This operation creates an Attachment entity.

Mandatory Attributes

Mandatory Attributes	Rule
@type	

Usage samples

Creating an Attachment

Request

```
POST /attachment
Content-Type: application/json

{
  "@type": "Attachment",
  "name": "Link to paperback",
  "url": "https://www.amazon.co.uk/Designing-Web-APIs-Building-
Developers/dp/1492026921",
```

© TM Forum 2026. All Rights Reserved.

```
  "validFor": {
    "startDateTime": "2024-11-05T10:15:26.007Z"
  },
  "attachmentType": "book",
  "mimeType": "text/html"
}
```

Response

```
201

{
  "@type": "Attachment",
  "href": "https://serverRoot/tmf-api/document/v5/attachment/1492026921",
  "id": "1492026921",
  "name": "Link to paperback",
  "url": "https://www.amazon.co.uk/Designing-Web-APIs-Building-
Developers/dp/1492026921",
  "validFor": {
    "startDateTime": "2024-11-05T10:15:26.007Z"
  },
  "attachmentType": "book",
  "mimeType": "text/html"
}
```

Updates partially a attachment

`PATCH /attachment/{id}?fields=…`

Description

This operation allows partial updates of an Attachment entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
href	
id	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

© TM Forum 2026. All Rights Reserved.

Patchable Attributes	Rule
attachmentType	
content	
description	
mimeType	
name	
size	
url	
validFor	

Usage samples

Here is an example of a request for patching an attachment.

Request

```
PATCH /attachment/1492026921
Content-Type: application/json

{
  "@type": "Attachment",
  "name": "Zelda as paperback"
}
```

Response

```
200

{
  "@type": "Attachment",
  "href": "https://serverRoot/tmf-api/document/v5/attachment/1492026921",
  "id": "1492026921",
  "name": "Zelda as paperback",
  "url": "https://www.amazon.co.uk/Designing-Web-APIs-Building-
Developers/dp/1492026921",
  "validFor": {
    "startDateTime": "2024-11-05T10:15:26.007Z"
  },
  "attachmentType": "book",
  "mimeType": "text/html"
}
```

Here is an example of a request for patching an attachment.

© TM Forum 2026. All Rights Reserved.

Request

```
PATCH /attachment/1492026921
Content-Type: application/merge-patch+json

{
  "@type": "Attachment",
  "name": "Zelda as paperback"
}
```

Response

```
200

{
  "@type": "Attachment",
  "href": "https://serverRoot/tmf-api/document/v5/attachment/1492026921",
  "id": "1492026921",
  "name": "Zelda as paperback",
  "url": "https://www.amazon.co.uk/Designing-Web-APIs-Building-
Developers/dp/1492026921",
  "validFor": {
    "startDateTime": "2024-11-05T10:15:26.007Z"
  },
  "attachmentType": "book",
  "mimeType": "text/html"
}
```

Here is an example of a request for patching an attachment.

Request

```
PATCH /attachment/1492026921
Content-Type: application/json-patch+json

[
  {
    "op": "replace",
    "path": "/name",
    "value": "Zelda as paperback"
  },
  {
    "op": "replace",
    "path": "/validFor",
    "value": {
      "startDateTime": "2023-11-05T10:15:26.007Z"
    }
  }
]
```

Response

© TM Forum 2026. All Rights Reserved.

```
200

{
  "@type": "Attachment",
  "href": "https://serverRoot/tmf-api/document/v5/attachment/1492026921",
  "id": "1492026921",
  "name": "Zelda as paperback",
  "url": "https://www.amazon.co.uk/Designing-Web-APIs-Building-
Developers/dp/1492026921",
  "validFor": {
    "startDateTime": "2023-11-05T10:15:26.007Z"
  },
  "attachmentType": "book",
  "mimeType": "text/html"
}
```

Here is an example of a request for patching an attachment.

Request

```
PATCH /attachment/1492026921
Content-Type: application/json-patch-query+json

[
  {
    "op": "replace",
    "path": "/name",
    "value": "Zelda as paperback"
  }
]
```

Response

```
200

{
  "@type": "Attachment",
  "href": "https://serverRoot/tmf-api/document/v5/attachment/1492026921",
  "id": "1492026921",
  "name": "Zelda as paperback",
  "url": "https://www.amazon.co.uk/Designing-Web-APIs-Building-
Developers/dp/1492026921",
  "validFor": {
    "startDateTime": "2024-11-05T10:15:26.007Z"
  },
  "attachmentType": "book",
  "mimeType": "text/html"
}
```

© TM Forum 2026. All Rights Reserved.

Deletes a attachment

`DELETE /attachment/{id}?{filtering}`

Description

This operation deletes an Attachment entity.

© TM Forum 2026. All Rights Reserved.

API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42
```

© TM Forum 2026. All Rights Reserved.

```
{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

© TM Forum 2026. All Rights Reserved.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

© TM Forum 2026. All Rights Reserved.

Acknowledgements

Release History

Release Number	Date	Release led by:	Description
Release 1.0	04/15/2017	Pierre Gauthier TM Forum Mariano Belaunde Orange Labs	First Release of the Document.
Release 2.0	11/06/2018	Mariano Belaunde Orange Labs	Alignment with Guidelines 3.0
Release 4.0	11/09/2021	Stephen Harrop Vodafone Group	Major update to accommodate patterns from TMF-API Design Guidelines v4.0 and schema library-based generation.
Release 5.0	10/01/2024	Uwe Sülter Sergei Lukin Deutsche Telekom	Migration to v5
Release 5.0	14 Oct 2024	Uwe Sülter Sergei Lukin Deutsche Telekom	release
Release 5.0	11/05/2026	Maciej Gancarz Comarch SA	Release 5 of the Conformance Profile

Contributors to Document

Pierre Gauthier	TM Forum
Mariano Belaunde	Orange Labs
Stephen Harrop	Vodafone Group
Uwe Sülter	Deutsche Telekom
Sergei Lukin	Deutsche Telekom
Maciej Gancarz	Comarch SA

© TM Forum 2026. All Rights Reserved.